Chapter 20 Review

Campbell Biology, AP Edition, 12e

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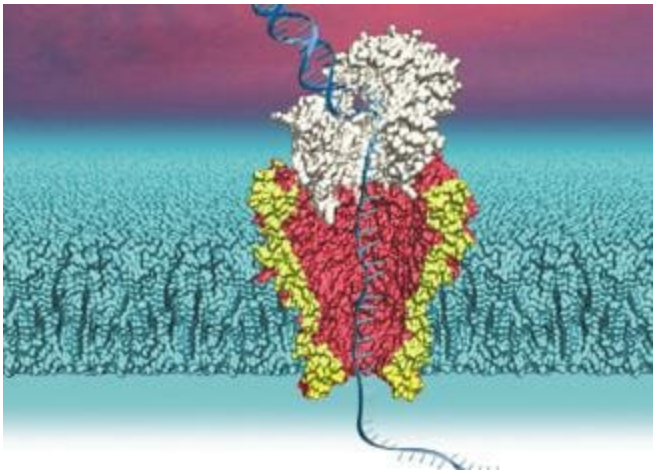
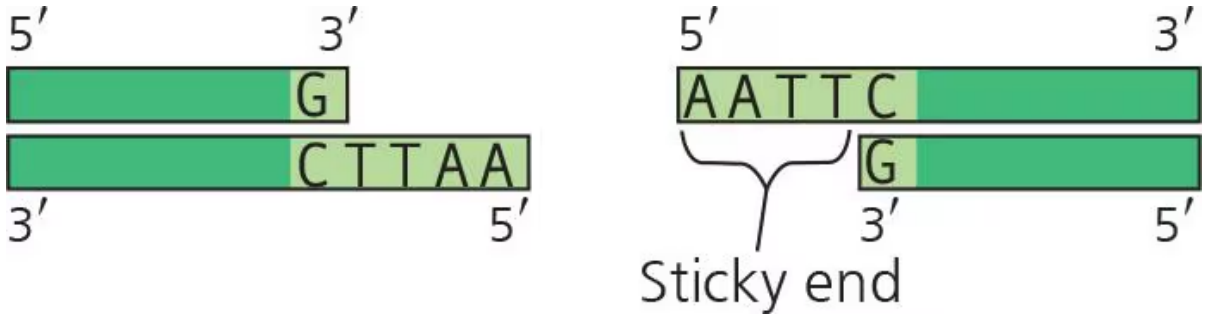


Summary of Key Concepts

In bacterial transformation experiments, plasmids are typically constructed with a gene for antibiotic resistance. Explain why that is important. Explain why in some environments there is a selective advantage for bacteria with the plasmid, but in other environments, there is a selective disadvantage. **(BIG IDEAS 1 & 3)**

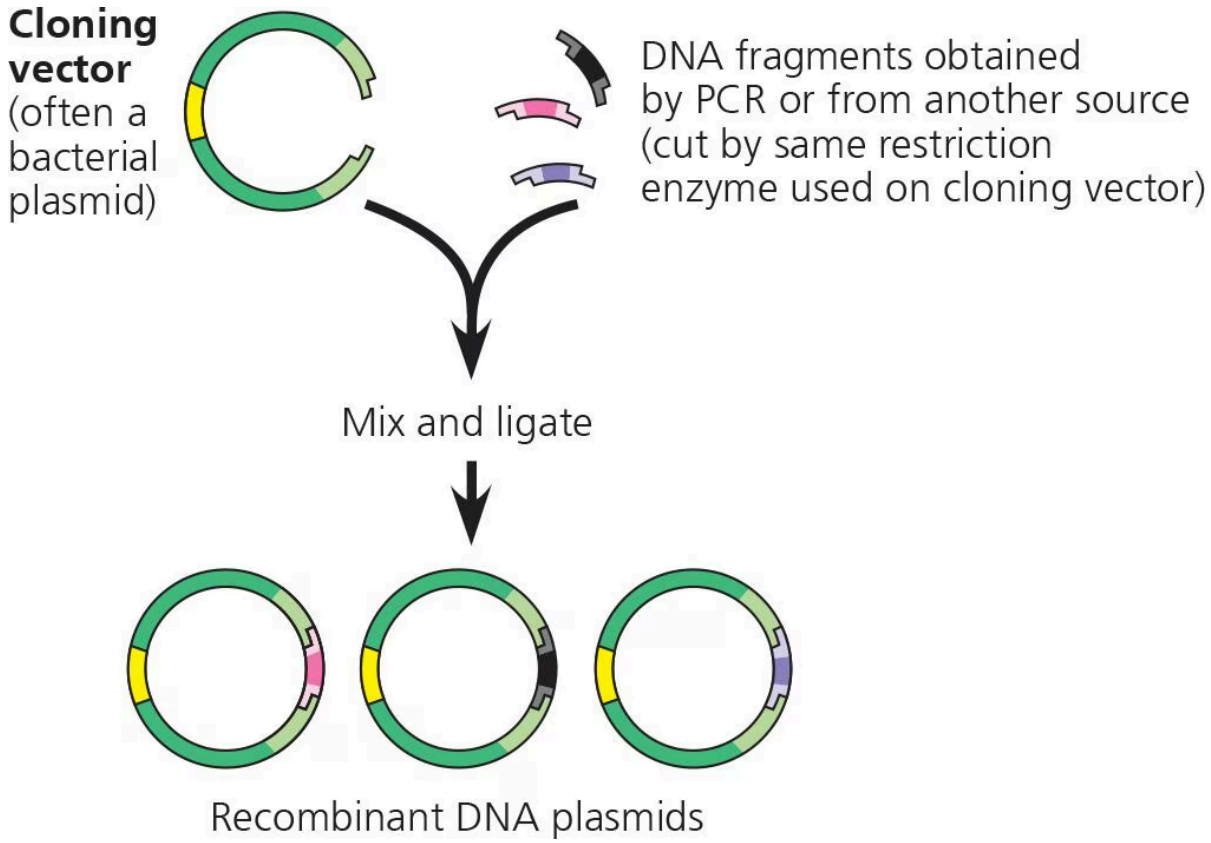
Concept 20.1: DNA sequencing and DNA cloning are valuable tools for genetic engineering and biological inquiry

- **Nucleic acid hybridization**, the base pairing of one strand of a nucleic acid to the complementary sequence on a strand from another nucleic acid molecule, is widely used in ****DNA technology**.
- ****DNA sequencing** can be carried out using the dideoxy sequencing method in automated sequencing machines.
- Fast and inexpensive next-generation (high-throughput) techniques for sequencing DNA are based on sequencing by synthesis: DNA polymerase is used to synthesize a stretch of DNA using a single-stranded template, and the order in which nucleotides are added reveals the sequence. Third-generation sequencing methods, including nanopore technology, sequence long DNA molecules one at a time.
- ****Gene cloning** (or ****DNA cloning**) produces multiple copies of a gene (or DNA segment) that can be used to manipulate and analyze DNA and to produce useful new products or organisms with beneficial traits.
- In ****genetic engineering**, bacterial ****restriction enzymes** are used to cut DNA molecules within short, specific nucleotide sequences (****restriction sites**), yielding a set of double-stranded ****restriction fragments** with single-stranded ****sticky ends**:



First fragment: Top strand: 5 prime, unlabeled region, G 3 prime. Bottom strand: 3 prime, unlabeled region, C T T A A 5 prime. (Sticky end). Second fragment: Top strand: 5 prime A A T T C (sticky end), unlabeled region, 3 prime. Bottom strand: 3 prime G, unlabeled region 5 prime.

- The sticky ends on restriction fragments from one DNA source can base-pair with complementary sticky ends on fragments from other DNA molecules. Sealing the base-paired fragments with DNA ligase produces ****__recombinant DNA molecules__.
- DNA restriction fragments of different lengths can be separated by ****__gel electrophoresis__.
- The ****__polymerase chain reaction (PCR)__ can amplify (produce many copies of) a specific target segment of DNA, using primers that bracket the desired sequence and a heat-resistant DNA polymerase.
- To clone a eukaryotic gene:



Cloning vector (often a bacterial plasmid) and DNA fragments obtained by PCR or from another source (cut by same restriction enzyme used on cloning vector) are mixed and ligated to form recombinant DNA plasmids.

- Recombinant ****__plasmids__ are returned to host cells, each of which divides to form a clone of cells.
- Expressing cloned eukaryotic genes in bacterial host cells poses several technical difficulties. The use of cultured eukaryotic cells as host cells, coupled with appropriate ****__expression vectors__, helps avoid these problems.

Describe how the process of gene cloning results in a cell clone containing a recombinant plasmid.

Concept 20.2: Biologists use DNA technology to study gene expression and function

- Several techniques use hybridization of a ****__nucleic acid probe__ to detect the presence of specific mRNAs.
- ***In situ*__hybridization__** and ****__RT-PCR__ can detect the presence of a given mRNA in a tissue or an RNA sample, respectively.
- Sets of genes co-expressed by a group of cells can be detected by ******RNA sequencing (RNA-seq)**—sequencing the **cDNAs** corresponding to mRNAs from the cells. ****__DNA microarrays__ are also used for this purpose.
- For a gene of unknown function, experimental inactivation of the gene (a gene knockout) and observation of the resulting phenotypic effects can provide clues to its function. The CRISPR-Cas9 system allows researchers to edit genes in living cells in a specific, desired way. The new alleles can be altered so that they are inherited in a biased way through a population (****__gene drive__).
- In humans, ****__genome-wide association studies__ identify and use ****__single nucleotide polymorphisms (SNPs)__ as genetic markers for alleles that are associated with particular conditions.

What useful information is obtained by detecting expression of specific genes?

Concept 20.3: Cloned organisms and stem cells are useful for basic research and other applications

- The question of whether all the cells in an organism have the same genome prompted the first attempts at organismal cloning.
- Single differentiated cells from plants are often ****__totipotent__: capable of generating all the tissues of a complete new plant.
- Transplantation of the nucleus from a differentiated animal cell into an enucleated egg can sometimes give rise to a new animal.
- Certain embryonic ****__stem cells__ (ES cells) from animal embryos and particular adult stem cells from adult tissues can reproduce and differentiate both in the lab and in the organism, offering the potential for medical use. ES cells are ****__pluripotent__ but difficult to acquire. Induced pluripotent stem (iPS) cells resemble ES cells in their capacity to differentiate; they can be generated by reprogramming differentiated cells. iPS cells hold promise for medical research and regenerative medicine.

Describe how, using mice, a researcher could carry out (1) organismal cloning, (2) production of ES cells, and (3) generation of iPS cells, focusing on how the cells are reprogrammed. (The procedures are basically the same in humans and mice.)

Concept 20.4: The practical applications of DNA-based biotechnology affect our lives in many ways

- DNA technology, including the analysis of genetic markers such as SNPs, is increasingly being used in the diagnosis of genetic and other diseases and in personal genome analysis. Personalized medicine offers potential for an individual minimizing their known risk for a disease, as well as better treatment of genetic disorders or cancers. ******Gene therapy** or gene editing with the CRISPR-Cas9 system may also lead to permanent cures. DNA technology is used with cell cultures in the large-scale production of protein hormones and other proteins with therapeutic uses. Some therapeutic proteins are being produced in ******transgenic** "pharm" animals.
- Analysis of genetic markers such as ******short tandem repeats (STRs)** in DNA isolated from tissue or body fluids found at crime scenes leads to a ******genetic profile**. Use of genetic profiles can provide definitive evidence that a suspect is innocent or strong evidence of guilt. Such analysis is also useful in parenthood disputes and in identifying the remains of crime victims.
- Genetically engineered microorganisms can be used to extract minerals from the environment or degrade various types of toxic waste materials.
- The aims of developing transgenic plants and animals are to improve agricultural productivity and food quality.
- The potential benefits of genetic engineering must be carefully weighed against the potential for harm to humans or the environment.

What factors affect whether a given genetic disease would be a good candidate for successful gene therapy?